

# Fahimeh Orvati Nia

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## Education

**Ph.D., Computer Engineering** (In Progress) *Jan 2025 – Present*

Texas A&M University, College Station, TX

Research Areas: AI, Computer Vision, Agricultural Imaging, Graph Neural Networks

Affiliated with Texas A&M AgriLife Research Institute

**M.Sc., Electrical Engineering** *Aug 2023 – Jan 2025*

University of Notre Dame, IN, USA

GPA: 3.74

Thesis: Time-series anomaly detection using GANs and pedestrian intention prediction with GNNs

**B.Sc., Electrical Engineering** *Sept 2016 – Jan 2020*

Amirkabir University of Technology, Tehran, Iran

## Experience

**Research Assistant, Texas A&M AgriLife Research** *Jan 2025 – Present*

- Computer vision and AI-based analysis of agricultural multispectral imagery.
- Plant growth analysis and phenotyping using deep learning.
- Integrating multispectral imaging with low-field MRI for comprehensive crop analysis.

**Teaching Assistant, University of Notre Dame** *Fall 2023 – Spring 2024*

- Courses: Introduction to Electrical Engineering; Signals and Systems.
- Developed course materials and supported students with assignments and projects.

**Research Assistant, University of Tehran** *Jan 2022 – June 2023*

- Machine learning for classification and clustering of musical instruments.

**IoT Developer, NOICT Company (INRA)** *July 2021 – Oct 2021*

- Wireless communication between IoT devices using the Zigbee protocol.

**Teaching Assistant, Amirkabir University of Technology** *Sept 2020*

- Guided students in designing control systems using MATLAB/Simulink.

## Selected Projects

**Traffic-Aware Pedestrian Intention Prediction** *2025*

Traffic-aware graph neural network architecture (TA-STGCN) for pedestrian intention prediction in dynamic traffic environments; evaluated on PIE dataset.

**Agricultural Imaging and Phenotyping** *2025 – Present*

Computer vision and AI methods for analyzing multispectral imagery of horticultural crops; integrating multispectral imaging with low-field MRI for comprehensive phenotyping.

**GAN-Based Anomaly Detection** *2024*

Designed GAN-based models with custom loss functions for time-series anomaly detection with applications in smart meter systems and power grids.

**Neighborhood Feature Aggregation for Remote Sensing** *2025*

Proposed neighborhood-based feature aggregation (pooling) for remote sensing image classification, targeting agricultural and environmental monitoring.

## Selected Publications

- **Fahimeh Orvati Nia**, Hai Lin. *Traffic-aware pedestrian intention prediction*. American Control Conference (ACC), 2025. arXiv:2507.12433  
Proposed a Traffic-Aware Spatio-Temporal GCN (TA-STGCN) that integrates dynamic traffic-signal states and contextual cues for pedestrian intention prediction. Achieved improved accuracy on the PIE dataset (4.75% over a baseline).
- **Fahimeh Orvati Nia**, Joshua Peeples, et al. *Neighborhood Feature Pooling for Remote Sensing Image Classification*. WACV 2026 Workshop on Earth Observation. arXiv:2510.25077  
Introduced a neighborhood feature pooling layer to capture local relationships and aggregate texture-like similarities across feature dimensions. Demonstrated consistent accuracy gains across datasets and backbones with minimal parameter overhead.
- **Fahimeh Orvati Nia**, S. Salehi, J. Peeples. *Evaluating GAN-LSTM for Smart Meter Anomaly Detection in Power Systems*. TPEC 2026. arXiv:2601.09701  
Conducted a power-system-oriented evaluation of GAN-LSTM anomaly detection on AMI time-series using the LEAD dataset (406 buildings, 1 year). Standardized preprocessing/windowing/thresholding and showed strong detection performance ( $F1 = 0.89$ ) versus common baselines.
- A. McFarland, L. Rossi, F. Nia, J. Peeples, A. Svyaneck. *Integrating Multispectral Imaging and Low-Field Magnetic Resonance Imaging for Comprehensive Phenotyping of Horticultural Crops*. HORTSCIENCE 60 (7), 2025.

## Skills

**Languages & APIs:** Python, C++, MATLAB, LaTeX

**ML & CV Libraries:** PyTorch, TensorFlow, Keras, OpenCV

**Developer Tools:** Git, Docker

**Embedded Systems:** STM32, Raspberry Pi, Arduino

## Activities and Service

**Peer Review:** LXAI-NeurIPS 2025; IEEE Robotics and Automation Letters (RA-L); ICRA 2026; ACC 2025; CCTA 2025

**Student Mentoring:** Chloe Tiley; Sophia Martinez-Badillo; Michael Morse; Amran Kassaye; Nazar Oladepo

## Awards and Scholarships

**Top rank in Iranian National Entrance Exam for Master's programs**

(Awarded)

**PlantCV Workshop + NAPPN 2026 Travel Award**

2026